MARMARA UNIVERSITY

FACULTY OF ENGINEERING

CSE3055-DATABASE SYSTEMS

(Fall 2025)

Step-II

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1.A. Entities and Their Definitions

1. **Customer**
   Customers are the individuals or companies who place orders for the products manufactured by Dene Technology. Based on these orders, the company initiates the production process. The database stores all relevant customer information.

   Attributes: CustomerID, Name, Address, Mail, PhoneNumber, RegistrationDate, CustomerType

2. **Employee**
   Employees are the personnel working in the Operations and Production Departments of Dene Technology. They are responsible for managing, supervising, and performing daily operational and manufacturing tasks.
   Attributes: EmployeeID, FullName(Name , Surname), Position, Salary, Mail, PhoneNumber, Address, HiredDate

3. **Product**
   Products represent the items ordered by customers and manufactured by the company, either independently or through collaboration with partners.
   Attributes: ProductID, Name, UnitPrice, Category, Cost, Stock, ManufacturedDate

4. **Payment**
   The Payment entity represents financial transactions related to the company's business activities. It includes payments received from customers for products, payments made to partner companies in joint projects, and payments made to suppliers for externally sourced parts.
   Attributes: PaymentID, PaymentType, PaymentDate, PaymentAmount, SwiftNo, PaymentMethod, ApprovedBy, Payee

5. **Invoice**
   Invoices are issued for three main purposes: payments received from customers, payments made to partner companies in collaborative productions, and payments made to suppliers for purchased parts.
   Attributes: InvoiceID, InvoiceDate, TotalAmount, TaxRate, TaxOfficeNumber, InvoiceType, InvoicingParty, InvoicedParty

6. **Supplier**
   Suppliers are external companies that provide the parts and components used in the production of Dene Technology ‘s products when such parts are not produced in-house.
   Attributes: SupplierID, Name, Address, Mail, PhoneNumber, ContactPerson, Country, Rating, PartnershipDate, ReceivedProduct

7. **Partner**
   Partners are companies that collaborate with Dene Technology in the joint production or development of certain products, typically within customer projects.
   Attributes: PartnerID, PartnerName, PartnerType, Country, PartnershipDate, Mail, PhoneNumber

8. **Machine**
   The Machine entity records information about all production machines owned by the company. It includes details about their status, maintenance history, and production capacity.
   Attributes: MachineID, MachineName, PurchaseDate, Status, Location, LastMaintenance, NextMaintenance, Capacity

9. **QualityCheck**

   The QualityCheck entity tracks the quality control activities conducted on products and components. It records who performed the check, which machine and product were involved, and the inspection result.
   Attributes: QualityCheckID, CheckDate, Result, CheckedBy

1.B. Business processes, rules and more

The main business process in Dene Technology starts when a customer places an order.

After the order is received, the production process begins. The company decides whether the product will be produced in-house using company machines or with the help of partner companies. If some parts are not available inside the company, they are supplied from external suppliers. Each step of the process is recorded in the database for tracking and control.

During the production phase, every product and machine used is registered in the system. When production is completed, the product goes through a quality check. The quality control process ensures that the product meets company standards. The result of each check is stored in the database and linked to the product, machine, and employee who performed the test.

After the product passes the quality check, it is delivered to the customer. Then, an invoice is

created and sent to the customer. Payments are tracked in the system — both the money received from customers and the money paid to suppliers or partners. All financial operations must be approved by an authorized employee.

1. **Product Rules**

- Each product must be requested by at least one customer
- Each product must have exactly one invoice
- A product may have zero or more suppliers, depending on whether it includes external parts
- A product may have zero or more partners, depending on collaboration or joint production
- Each product must be assigned to at least one employee who works on it
- Each product must have at least one payment, since it is sold or produced for a cost
- Each product must be produced using at least one machine owned by Dene Technology
- Each product must have at least one quality check record

2. **Customer Rules**

- Each customer must have at least one product order
- Each customer must have at least one payment (every registered customer must have made a transaction)
- Each customer must have at least one invoice

3. **Employee Rules**

- Each employee must work on at least one product
- Each employee may have a supervisor
- Each employee may be linked with one or more suppliers or partners, depending on their job duties
- Each employee may perform one or more quality checks, but this is optional

4. **Payment Rules**

- Each payment may be linked to a customer (if it is a sale), but this is optional
- Each payment must be related to exactly one product
- Each payment must be related to exactly one invoice
- Each payment may be linked to one or more suppliers or partners
- Invoice Rules
- Each invoice must be related to exactly one product
- Each invoice may be related to one customer, supplier, or partner, depending on the transaction type

5. **Invoice Rules**

- Each invoice must be related to exactly one product b. Each invoice may be related to one customer, supplier, or partner, depending on the transaction type

6. **Partner Rules**

- Each partner must be connected to the company through at least one employee
- Each partner must have one or more payments
- Each partner must have worked on at least one product
- Each partner must have at least one invoice

7. **Supplier Rules**

- Each supplier must be connected to the company through at least one employee
- Each supplier must have one or more payments
- Each supplier must have worked on at least one product
- Each supplier must have at least one invoice

8. **Machine Rules**

- Each machine must be linked to exactly one product, because each product or part is produced by a specific machine

9. **QualityCheck Rules**

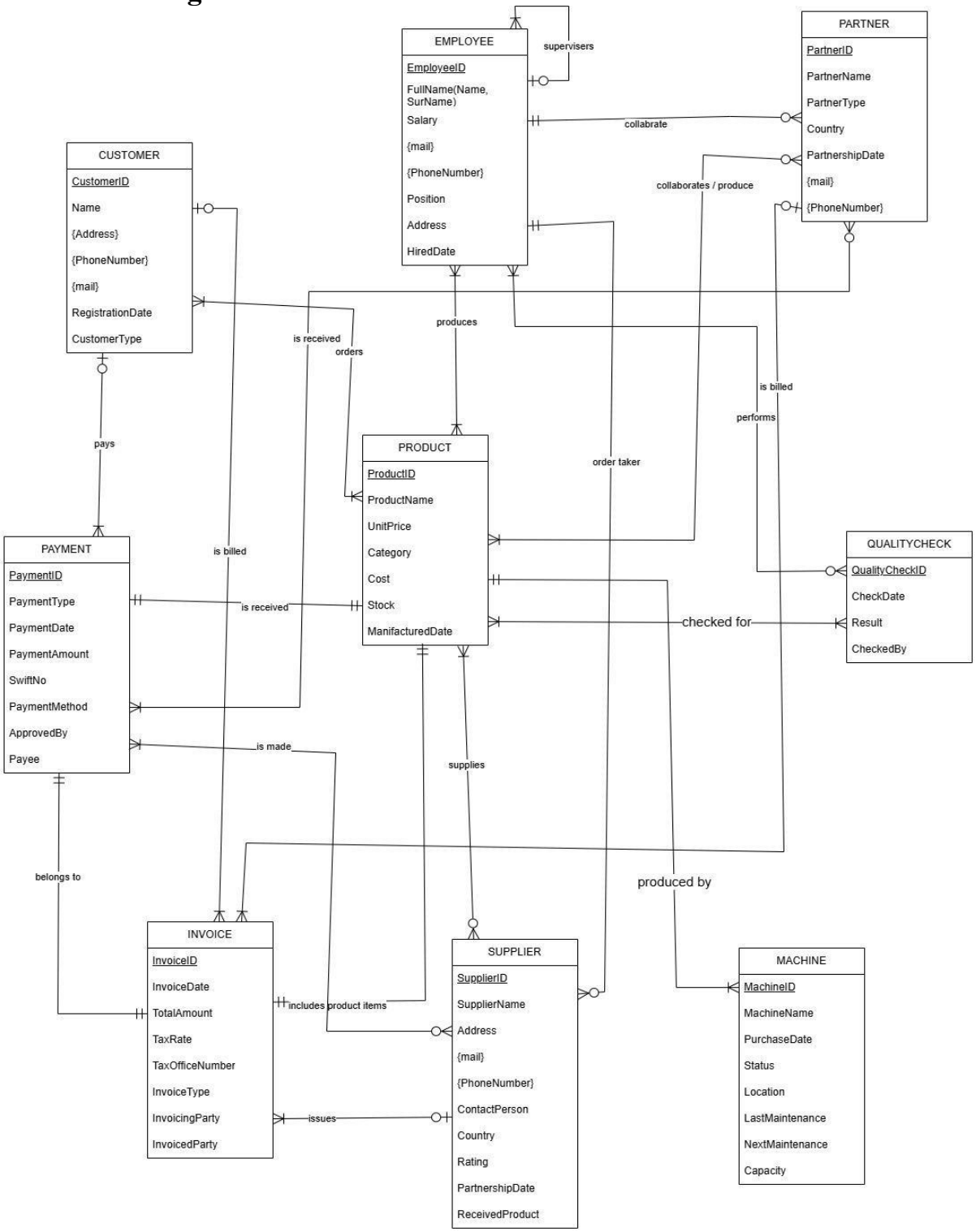
- Each quality check must be performed by exactly one employee
- Each quality check must be related to at least one product, because it exists only for product inspection

1.C. Other Functional & Non-Functional Business Requirements

Functional Requirements

- The system must allow the registration of new customers, employees, products, suppliers, and partners
- Developers must be able to create, update, and delete records for products, invoices, and payments
- The database must record all relationships between entities such as products, machines, and quality checks
- The system must allow authorized employees to approve payments and invoices
- The system must provide a simple web interface for data entry and management

Non-Functional Requirements

- The system must ensure data consistency and accuracy across all tables
- Only authorized users can access or modify data
- The database must be designed to handle future growth in the number of customers and products

2. ER/EER Diagram